

STATEMENT OF WORK (SOW)

TASK TITLE: Large Scale Torus Strap Indentation Testing – Series 2

1.0 INTRODUCTION/BACKGROUND

NASA is developing Hypersonic Inflatable Aerodynamic Decelerators (HIADs) for application to future robotic and human missions requiring atmospheric entry. HIADs offer considerable system mass and volume savings over rigid aeroshell technology. In addition, inflatable aeroshells are not constrained by the launch vehicle payload shroud diameter, which offers the possibility to lower ballistic coefficient of the entry system while delivering a larger payload mass fraction than a rigid aeroshell counterpart.

NASA Langley Research Center (LaRC) led the successful flight test of a 6-meter diameter HIAD in November 2022 under the project called the Low earth Orbit Flight Test of an Inflatable Decelerator (LOFTID) project. NASA is currently developing a 10 meter diameter HIAD design for an engine reuse application. The purpose of this task is to conduct strap indentation testing of torus test article from the 10m HIAD design.

The Statement of Work (SOW) defines the work to be performed by the Contractor to conduct strap indentation testing on **one (1) torus** and provide a test report and data.

2.0 SCOPE OF WORK

2.1 SCOPE AND GENERAL TASKS

The Contractor shall provide all resources, including but not limited to, test facilities, test equipment, test operations, documentation, and delivery of test data and reports as defined in this SOW. The Contractor's scope of work shall include the general tasks listed below. The specific requirements for these tasks are provided in SOW sections that follow.

- 1) Participate in Kick-off Meeting
- 2) Torus Test Procedure Development
- 3) Torus Test Operations (1 torus)
- 4) Test Report and Data Delivery

2.2 MEETINGS AND REVIEWS

The Contractor shall conduct and/or support the meetings and reviews defined below.

2.2.1 Kick-off Meeting. The Contractor shall attend a Kick-off Meeting that will be held via telephone conference and web conferencing. The Kick-off Meeting will provide a forum at which the Government and the Contractor will review the test requirements described in this SOW. The Kick-off Meeting will be held using web conferencing.

2.2.2 Test Readiness Review. The Contractor shall conduct a Test Readiness Review. See paragraph 4.2 for requirements for this review. The review will be held using web conferencing.

2.2.3 Additional Meetings. Additional meetings will be held to support the test activities as needed. The Contractor shall contact the Government if needed to support the test planning, preparations, or discuss questions and issues. Any additional meetings that are needed will be held using web conferencing.

3.0 TEST ARTICLE

The inflatable torus test article to be tested will be provided to the Contractor as Government Furnished Property (GFP). Figures 1 and 2 provide the outer dimensions and cross section

of the torus test article. The torus is constructed with 30-inch diameter Zylon braid material to provide shape and strength. A multi-ply Teflon liner provides the gas barrier to maintain the pressure and minimize leakage. Other features include four (4) axial cords, four (4) port/line assemblies for inflation and venting, and an outer RTV coating. Figure three shows an example of an inflation port during fabrication and completed port with an inflation hose attached. Each inflation line will be terminated with a $\frac{1}{4}$ turn ball valve and quick disconnect fitting for easy attachment. The torus article to be tested is a **V1.0 torus design with a MEOP of 36 psi, and includes twenty (20) straps** for indentation testing. The V1.0 torus has been pressure tested to 45psi by the torus manufacturer to demonstrate a safety factor of 1.2x. It should be noted that the torus to be tested has already gone through strap indentation testing at the Contractor's facility and will be reused with newly applied straps for this effort.

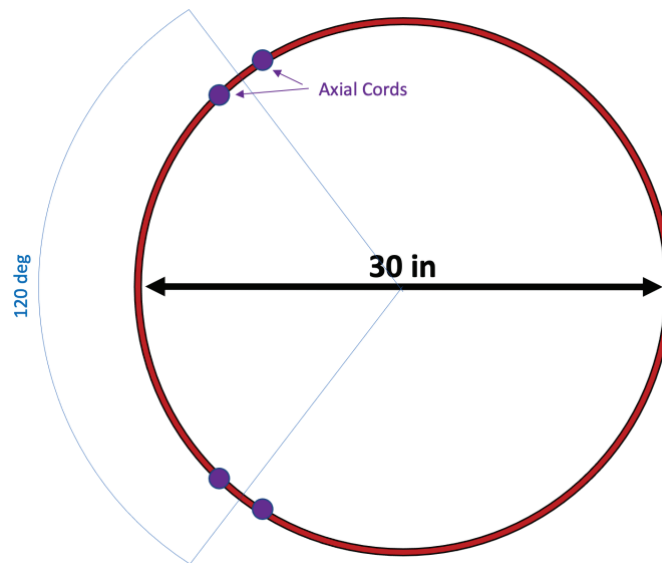


Figure 1. Torus Cross-Section

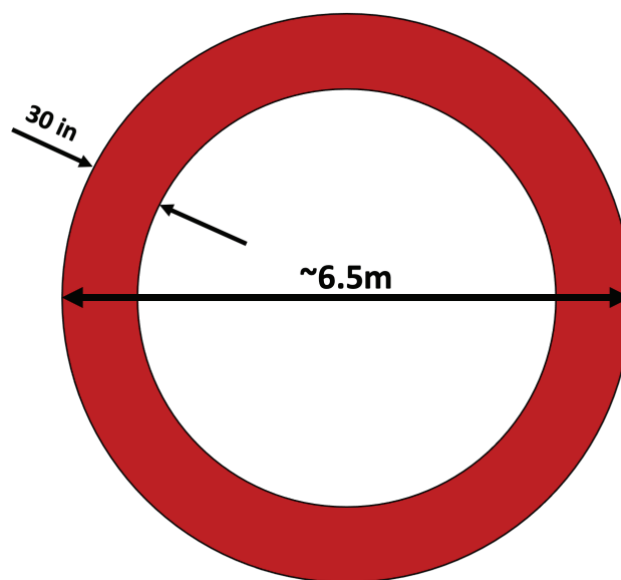


Figure 2. Torus with Dimensions

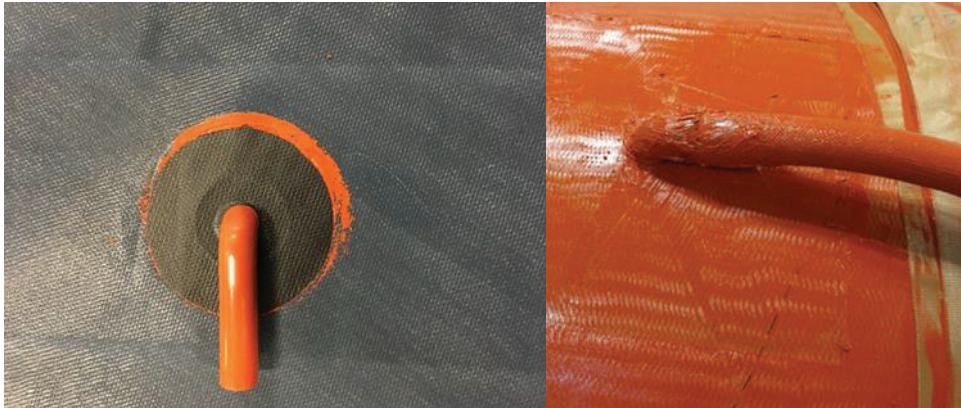


Figure 3. Example Port and Fill Line Installation

4 STRAP INDENTATION TESTING

The objective of this task is to determine the load carrying capacity of the structural webbing to torus interface, and the resulting indentation of the torus. The testing will be done at multiple torus internal pressures and using multiple different webbing widths (2", 3" and possibly 4"). The Contractor shall provide all services, facilities, and requirements necessary to conduct the strap indentation test as defined below.

- 4.1** The Contractor shall utilize the test plan and procedure (one document) that was generated to perform the strap indentation testing in 2024.
- 4.2** The Contractor shall conduct a Test Readiness Review that is presented to the Government at least three days prior to conducting the test. The objective of the Test Readiness Review is to clearly show that all test preparations have been completed and the test articles, procedures, equipment, and safety features are complete and ready to conduct the test. The Test Readiness Review shall include description of the test articles, test requirements, test configuration, test procedures, test equipment and instrumentation, data collection plan, test team and responsibilities, safety assessment, emergency and safety procedures, and plan for post-test documentation. Any major issues or actions from the Test Readiness Review shall be addressed prior to conducting the test.
- 4.3** The test configuration and procedures shall include the following for testing one torus at a time (3 total torus tests):
 - 4.3.1** The torus shall be positioned such that the dual load applied to each of the two strap ends is at a 25 deg angle from the plane of the torus.
 - 4.3.2** The two load application devices shall be in the same plane spaced 30" apart.
 - 4.3.3** Two reacting straps or slings (GFE) shall provide the anchor points for the torus to react the load applied by the load application devices.
 - 4.3.4** The torus shall be prevented from rotating during loading.
 - 4.3.5** The load in each load application device shall be monitored with a calibrated load cell.
 - 4.3.6** The torus shall be monitored with photogrammetry during testing to measure the indentation of the torus to an accuracy of ± 0.05 ".
 - 4.3.7** **Photogrammetry shall be performed with 3 ARAMIS Systems to get views of all torus surfaces of interest. The Contractor shall perform initial photogrammetry data reduction.**
 - 4.3.8** The torus shall include multiple straps at different radial locations, requiring the torus to be rotated within the test fixture as each of the tests are performed.
 - 4.3.9** Testing shall be conducted using load control with continuous photogrammetry data unless otherwise specified by the government. The load control shall have a target load provided by the Government, with the test conductor and technical representative able to call an end to the test early if load carrying capacity is no long increasing.
 - 4.3.10** The torus shall be tested at multiple internal pressures (with an upper limit of 36psi)

specified by the Government equal to or less than MEOP of the torus design.

- 4.3.11** The test fixture's load application devices shall be able to perform asymmetric loading (e.g. 3000lbs on each strap, or 4000lbs on one strap and 1000lbs on the second).
- 4.3.12** **The test fixture shall use lightweight shackles for attachment to the torus webbing that are 4.25" wide to allow for webbing up to 4" wide. The design shall be reviewed by the government before fabrication.**
- 4.3.13** **String Pots shall be used to measure the displacement of the load application devices during testing.**
- 4.3.14** A physical barrier shall be constructed for safety purposes in the event of a rapid depressurization of the torus during testing. Remote operation of testing maybe required to meet safety requirements.
- 4.3.15** NASA personnel shall be allowed to witness testing in person.
- 4.4** Test data shall be recorded throughout the test operation and shall include torus pressure (1 Hz) and loadcell data (1 Hz), photogrammetry data each test increment, and test time.
- 4.5** Torus pressure data and strap load shall be displayed real-time for viewing during testing at a rate of 1 Hz.
- 4.6** **4 cameras shall be recorded to a hard drive that capture three views of the torus during testing and one view of the data system display screen in the control room. Data shall be transferred to Hard Drive before delivery to the Government.**
- 4.7** After test completion, a test report shall be submitted that includes a description of the test article, test operations, test results, lessons learned, and all test data.

Table 1: Test Summary

Torus ID	Design	Maximum Operating Pressure	Straps	Test Pressure(s)
MDU3-R-1	V1.0	36 psi (pressure tested to 45 psi)	20	36 psi

5 GENERAL TASK REQUIREMENTS

- 5.1** The Contractor shall provide project management and coordination and work with NASA designated personnel to ensure communication of requirements and resolution of task issues.
- 5.2** The Contractor shall provide on-site access to the Government designated personnel to include engineering, management and quality assurance for oversight during this task.

6 DELIVERABLES

6.1 Documentation

The Contractor shall deliver documentation in electronic format (Microsoft Word or PowerPoint) to Steve Hughes at stephen.j.hughes@nasa.gov and Robert Mosher at robert.n.mosher@nasa.gov. The documentation shall include:

- a) Test Plan
- b) Test Procedure
- c) Test Reports including a description of the data analysis process and any information required to recreate this analysis in the future based on the raw images
- d) Test Data including:
 - i. torus pressure,
 - ii. load cell readings,
 - iii. string pot readings,
 - iv. raw photogrammetry images,
 - v. processed photogrammetry solutions including deflection vs. time for critical areas

7 SCHEDULE OF DELIVERABLES AND MILESTONES

7.1 Period of Performance

The period of performance for the Task shall be 70 calendar days after award.

7.2 Milestone and Deliverable Schedule

The Contractor shall provide the deliverables under this Task Order in accordance with the following milestone schedule.

Deliverables and Milestones

Item Number	Task Reference	Deliverable	Schedule
7.1	2.2.1	Kick-off Meeting	Within 5 business days after Award
7.2	2.2.2, 4.2	Test Readiness Review	At Least 3 business days prior to Test
7.3	4.3.12	Lightweight Shackle Design	Within 20 days after Award
7.4	4.5 6	Test Report and data	Within 30 business days after completion of Test

8 GOVERNMENT FURNISHED INFORMATION/PROPERTY

List of Government-Furnished Property Furnished Pursuant to 52.245-1 (NFS 1852.245-76) (JAN 2011)

For performance of work under this task order, the Government will make available Government property identified below on a no charge-for-use basis pursuant to the clause at FAR 52.245-1, Government Property, as incorporated in the basic contract. The Contractor shall use this property in the performance of this task order at the Contractor's facility(s) and at other location(s) as may be approved by the Contracting Officer. Under the FAR 52.245-1, the Contractor is accountable for the identified property.

Item No.	SOW Ref No.	Item Description	Approximate Quantity
8.1	3	Torus Test Article #1 (V1.0 Design)	1